Redbird Flight Simulations
Qualification and Approval Guide (QAG)

MDN Advanced Aviation Training Device
QAG
VERSION 0
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FAA-Approved QAG

Section 1: List of Configurations

1. **Configuration: PA46-500-E1-** Single Engine Turbo Prop, Land, Retractable Gear, Constant Speed Propeller, Low Wing

2. **Configuration: PA46-500-G1-** Single Engine Turbo Prop, Land, Retractable Gear, Constant Speed Propeller, Low Wing

FAA-Approved QAG

Statement of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b.:

The devices and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.
"This is to certify that Microsoft Corporation, the owner and developer of the ESP Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Section 2: Description of Hardware and Software Components

All Configurations Include:

Qty	Manufacturer	Configuration	Description/Function
1	Redbird Flight Simulations	All	Central Processor– Sim Platform Software
1	Redbird Flight Simulations	All	Central Processor– Cockpit Platform Software
1	Microsoft	All	Operating system
1	Microsoft	All	Simulation engine
1	Redbird Flight Simulations	All	Map, track, and glideslope tracking
6	22" Industry Standard	All See Fig- 13	Flat panel displays for exterior views
3	19" Industry Standard	All See Fig- 13	Flat panel displays for virtual instruments
1	Redbird Flight Simulations	All See Fig- 14	Cockpit enclosure and platform
1	Redbird Flight Simulations	All See Fig- 13	Dual rudder pedals
1	Redbird Flight Simulations	All See Fig- 13	Dual yoke control
1	Redbird Flight Simulations	All See Fig- 3, 5	Throttle quadrant
1	Redbird Flight Simulations	All See Fig- 1, 2	Virtual instrument controls overlay
1	Redbird Flight Simulations	All See Fig- 13	Lower panel switch block
1	Redbird Flight Simulations	All See Fig- 11	Left side circuit breaker panel
1	Redbird Flight Simulations	All See Fig- 12	Right side circuit breaker panel

Individual Configurations Include:

1. **PA46-500-E1**
   Throttle Quadrant: Single engine, turbo prop. See Figure- 3
   Virtual Instrument Controls Overlay: Redbird Entegra with 2 Primary Flight Displays (PFD) and 1 Multi-Function Display (MFD), dual Redbird 430s, and Redbird 1500 autopilot. See Figure- 1

2. **PA46-500-G1**
   Throttle Quadrant: Single engine, turbo prop with Flight Management System (FMS) keypad. See Figure- 4
   Virtual Instrument Controls Overlay: Redbird 1000 with 2 PFDs and 1 MFD, integrated Redbird 700 autopilot. See Figure- 2

Section 3: Statements of Compliance

All Configurations, as noted, meet AC 61-136, Appendix 2, Paragraph 8 BATD Design Criteria:

a. Controls:

(1) Meets Requirements: Physical flight and aircraft system controls are recognizable as to their function and how they are manipulated solely from their appearance. There is no keyboard, mouse, or gaming joystick inside the Redbird MDN Cockpit. See Figures- 1,2

(2) Meets Requirements: Virtual controls are limited to the initialization of the device and instructor input. Avionics are virtually displayed but are operated with physical buttons and knobs that closely represent the aircraft. See Figure- 13

(3) Meets Requirements: No keyboard or mouse is used to set or position any feature of the device. The required and additional equipment is operated in the same manner in which it would be operated in the represent aircraft.

(4) Meets Requirements: The arrangement, appearance, and operation of controls, instruments, and switches closely represent the aircraft, including: battery, alternators, fuel boost pumps, avionics master, pitot heat, and required lights. Magnetos are not applicable to the aircraft represented. See Figure- 13

(5) Meets Requirements: During start-up of the device the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and logging of the flight training data to the student key will not be allowed.

b. Control Requirements:

(1)(a): Meets Requirements: Dual self-centering displacement yokes that are interlinked and allow for continuous adjustment of pitch and bank. The yoke handles are aircraft style with electric trim, autopilot controls, a push to talk switch, and a transponder ident button as applicable to the aircraft. See Figures- 6, 7, 8, 9

(1)(b): Meets Requirements: Dual self-centering rudder pedals that are interlinked and allow for continuous adjustment of yaw. See Figure- 10

(1)(c): Meets Requirements: Power control lever that allows for continuous movement from idle to max power, with beta and reverse settings as applicable to the aircraft. See Figure-

(1)(d): Meets Requirements: Condition control lever, Power control lever, and Manual Override control lever as applicable to the aircraft. See Figure- 4

(1)(e): Meets Requirements: See Individual Configuration.

c. Control Inputs:

(1) Meets Requirements: The transport delay for all control inputs is 300 milliseconds or less. The required transport delay is tested at system start-up by internal start/stop timers. Redbird Flight Simulations is the manufacturer of all control inputs and uses USB 2.0 which exceeds this requirement. Redbird Flight Simulations certifies that the MDN meets this standard.

(2) Meets Requirements: Each control input is tested at training session start-up. If any device fails to meet the requirements of Appendix 2, Paragraph 8, Section c, item (1) a warning message is displayed and the device will not start a training session.

d. Display Requirements:

(1) Meets Requirements: See individual configuration.

(2) Meets Requirements: All instrument displays included in each individual configuration are visible during all flight operations. The displays are updated at rate that insures that the image of the instrument does not appear to: be out of focus, illegible, jump, step, jagged, nor is it distracting.

(3) Meets Requirements: Display updates exceed 60Hz. All replicated instruments display changes equal or greater than the values required by this item.

(4) Meets Requirements: Displays reflect the dynamic behavior of an actual aircraft in flight.

e. Flight Dynamics Requirements:

(1) Meets Requirements: All flight dynamics are comparable to the way the represented aircraft performs and handles.

(2) Meets Requirements: All simulated aircraft performance parameters are comparable to the aircraft being represented.

(3) Meets Requirements: Vertical lift component changes as a function of bank, comparable to the aircraft being represented.

(4) Meets Requirements: Changes in flap setting and gear position are accompanied by changes in flight dynamics, comparable to the aircraft being represented.

(5) Meets Requirements: The handling and performance qualities of the simulated aircraft are affected by the presence and intensity of wind and turbulence in manner comparable to the aircraft being represented.

f. Instructional Management Requirements:

(1) Meets Requirements: The instructor has the ability to pause the system at any time using either the buttons on the instructor's console or the buttons on the instrument panel.

(2) Meets Requirements: The instructor has the ability to manipulate the parameters listed in this item independently of the simulation.

(3) Meets Requirements: The system records both the horizontal and vertical track of the aircraft movement and this can be view in real-time or at a later time.

(4) Meets Requirements: The instructor is able to disable instruments prior to or during a training session and simulate failure of the instruments without stopping or freezing the simulation to affect the failure.

(5) Meets Requirements: The device includes a navigational data base that covers all of North America and is based on procedures as published per 14 CFR part 97.

All configurations, as noted meet AC 61-136, Appendix 3, Paragraph 3 AATD Design Criteria:

a.

(1) Meets Requirements: The device enclosure and cockpit design and instrument panel arrangement represent a Piper Malibu Meridian. See Figure- 13

(2) Meets Requirements: Cockpit knobs/system controls/switches/switch panels are realistic in size and design appropriate to each intended function, in the proper position and distance from the pilot's seated position, and representative of the category and class of the aircraft being represented. See individual configuration.

(3) Meets Requirements: Primary flight and navigation instruments are life size and exhibit neither stepping nor excessive transport delay. They are arranged so as to observe trends and provide a realistic scan pattern. See individual configuration.
(4) Meets Requirements: See individual configuration.
(5) Meets Requirements: See individual configuration.
(6) Meets Requirements: See individual configuration.
(7) Meets Requirements: Manual and electric pitch trim that move an electric trim indicator in a ratio representative of the aircraft. Electric pitch trim switches are located on both yokes and when activated move the manual trim well as appropriate to the aircraft represented. See Figure- 5, 7, 9
(8) Meets Requirements: All visual cues for both day and night, VFR and IFR are provided. See Figure- 13
(9) Meets Requirements: Instructors station is run from a "tablet" style or laptop PC from inside or outside the device. The instructor can oversee the pilot's horizontal and vertical flight profiles in real time and space. The instructor has the ability to complete all functions listed in this item without interrupting the flight. See Figure-
(10) Cockpit seat is fully adjustable.

Configuration: PA46-500-E1

This configuration represents a single engine turbo-prop, land, retractable gear, constant speed propeller, low wing, airplane representative of a Piper Malibu Meridian.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430s).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat- N/A by type of aircraft represented.
Cowl flaps- N/A by type of aircraft represented.
See Figures- 1, 3, 13

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Electronic Redbird Entegra with dual PFDs and one MFD with reversionary mode and back-up flight and navigation instruments. See Figure-
Sensitive altimeter that meets the requirements listed in this item. (Dual Redbird Entegra PFDs and back up altimeter)
Magnetic compass.
Heading indicator that meets the requirements listed in this item. (Dual Redbird Entegra PFDs and back up heading indicator)
Airspeed indicator that meets the requirements listed in this item. (Dual Redbird Entegra PFDs and back up heading indicator)
Vertical speed indicator that meets the requirements listed in this item. (Dual Redbird Entegra PFDs)
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. (Dual Redbird Entegra PFDs)
Slip and skid indicator that meets the requirements listed in this item. (Dual Redbird Entegra PFDs)
Engine instruments applicable to the aircraft being represented. (Redbird Entegra dual PFDs and one MFD)
Suction gauge applicable to the aircraft being represented. (Redbird Entegra MFD)
Flap setting indicator applicable to the aircraft being represented.

Pitch trim indicator that meets the requirements listed in this item. (Digital display on throttle quadrant)
Communication radio that meets the requirements listed in this item. (Dual Redbird 430s)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430s)
Transponder panel that meets the requirements listed in this item. (Redbird 330)
Fuel quantity indicators that meet the requirements listed in this item. (Redbird Entegra dual PFDs and one MFD)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Appropriate cockpit knobs/ system controls/switches/switch panels in realistic size and design to each intended fuction, in the proper position and distance from the pilot's seated position that are representative of the aircraft simulated. See Figure- 13

(3) Life sized Redbird Entegra with dual PFDs and one MFD with dual Redbird 430s GPS, communication and navigation radios, Redbird 1500 autopilot, Redbird 340 audio panel, Redbird 330 transponder, and back up heading indicator, attitude indicator, altimeter, and course deviation indicator representative of aircraft being simulated. Instruments exhibit neither stepping nor excessive transport delay, and are arranged so as to observe trends and provide a realistic scan pattern. See Figure- 1

(4) Redbird Entegra with dual PFDs and one MFD with dual Redbird 430s GPS, communication and navigation radios
See Figure- 1

(5) Dual Redbird 430s. See Figure-1

(6) Redbird 1500 autopilot and flight director displayed on dual Redbird Entegra PFDs as appropriate to represented aircraft. See Figure- 1

Configuration: PA46-500-G1

This configuration represents a single engine turbo-prop, land, retractable gear, constant speed propeller, low wing, airplane representative of a Piper Malibu Meridian.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat- N/A by type of aircraft represented.
Cowl flaps- N/A by type of aircraft represented.

See Figures- 2, 5, 13

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Electronic Redbird 1000 with dual PFDs and one MFD with reversionary mode and back-up flight and navigation instruments. See Figure- 2
Sensitive altimeter that meets the requirements listed in this item. (Dual Redbird 1000 PFDs and back up altimeter)
Magnetic compass.
Heading indicator that meets the requirements listed in this item. (Dual Redbird 1000 PFDs and back up heading indicator)
Airspeed indicator that meets the requirements listed in this item. (Dual Redbird 1000 PFDs and back up heading indicator)
Vertical speed indicator that meets the requirements listed in this item. (Dual Redbird 1000 PFDs)
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. (Dual Redbird 1000 PFDs)
Slip and skid indicator that meets the requirements listed in this item. (Dual Redbird 1000 PFDs)
Engine instruments applicable to the aircraft being represented. (Redbird 1000 dual PFDs and one MFD)
Suction gauge applicable to the aircraft being represented. (Redbird 1000 MFD)
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item. (Digital display on throttle quadrant)

Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)
Transponder panel that meets the requirements listed in this item. (Redbird 1000)
Fuel quantity indicators that meet the requirements listed in this item. (Redbird 1000 dual PFDs and one MFD)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Appropriate cockpit knobs/ system controls/switches/switch panels in realistic size and design to each intended fuction, in the proper position and distance from the pilot's seated position that are representative of the aircraft simulated.
See Figure- 13

(3) Life sized Redbird 1000 with dual PFDs and one MFD with Redbird 700 autopilot, Redbird 1000 FMS keypad, and back up heading indicator, attitude indicator, and altimeter representative of aircraft being simulated. Instruments exhibit neither stepping nor excessive transport delay, and are arranged so as to observe trends and provide a realistic scan pattern.
See Figure- 2

(4) Redbird 1000 with dual PFDs and one MFD with integrated GPS, communication and navigation radios
See Figure- 2

(5) Redbird 1000. See Figure- 2

(6) Redbird 700 autopilot and flight director displayed on dual Redbird 1000 PFDs as appropriate to represented aircraft. See Figure- 2

Section 4: Figures and Photos

Figure 1: PA46-500-E1 Instruments



Figure 2: PA46-500-G1 Instruments



Figure 3: Throttle Quadrant



Figure 4: Throttle Quadrant Detail



Figure 5: Throttle Quadrant with FMS Keypad

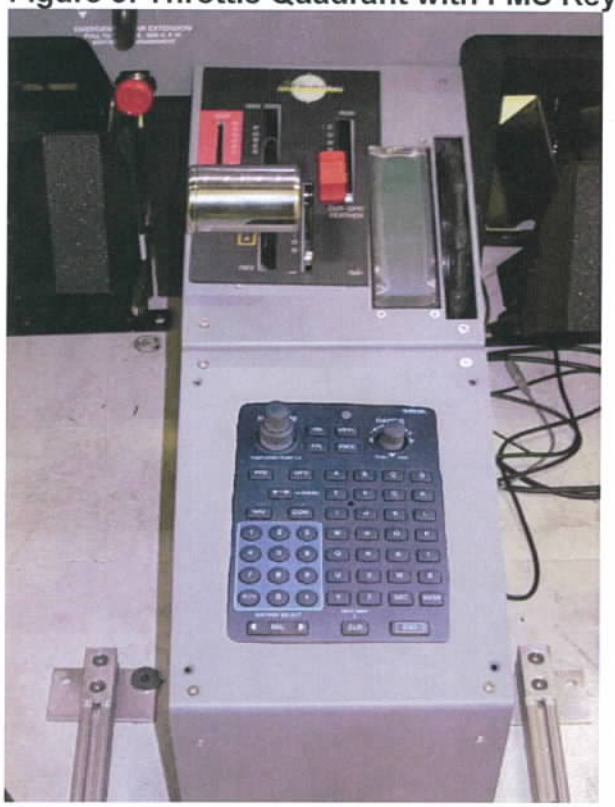


Figure 6: Pilot Yoke

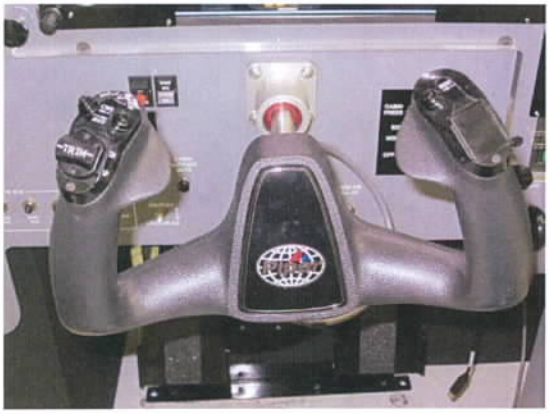


Figure 7: Pilot Side Switches



Figure 8: Co-Pilot Yoke



Figure 9: Co-Pilot Yoke Switches



Figure 10: Rudder Pedals Pilot and Co-Pilot

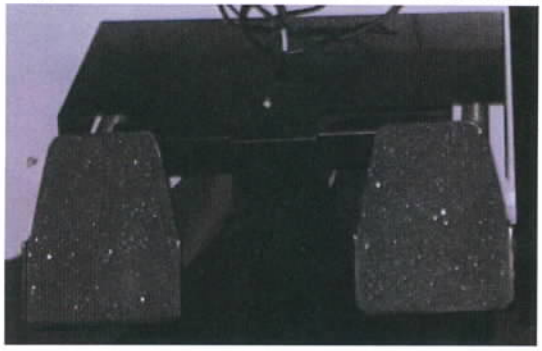


Figure 11: Circuit Breaker Panel Pilots Side



Figure 12: Circuit Breaker Panel Co-Pilots Side

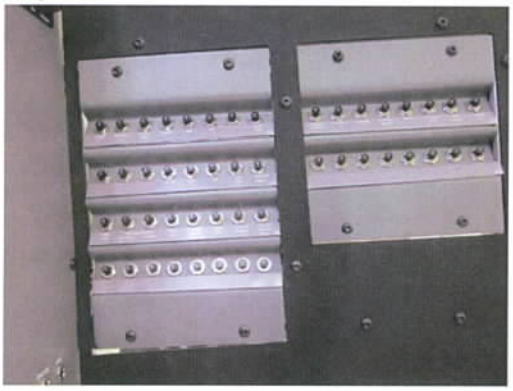


Figure 13: Assembled Cockpit



Figure 14: Motion Platform and Cockpit Enclosure

